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Structure and strain of phosphorus-vacancy clusters in highly doped silicon

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ABSTRACT

Ab initio calculations were used to assess the feasibility of P_nV ($n = 1$–4) cluster defects in highly phosphorus-doped silicon (Si:P). Unit cell lattice parameters were determined by density functional theory (DFT) and compared with published experimental high-resolution x-ray diffraction data. This comparison was uniquely enabled by applying pseudomorphic constraints directly into calculations. Across the ∼1.5%–9% P range considered here, lattice parameters for PV do not match experiment while, for P_3V and P_4V, they are physically consistent. P_2V shows experimental agreement at 3.3% P but not at higher 7.1% P. Investigating individual defects revealed PV and P_2V are not energetically favorable compared to keeping phosphorus isolated and electronically active. Conversely, P_3V and P_4V clustering have no energy penalty further signifying their viability. Detailed structural atomic analysis shows that these two higher-order clusters negate tensile strain of an otherwise isolated vacancy, attributed to charge density from absent phosphorus bonds. The transition from P_3V to P_4V lacks a strong thermodynamic driving force and, using kinetic Monte Carlo simulations, is argued to be governed by dopant availability instead. DFT also indicates the possibility of P_3V, P_4V, active phosphorus, and V^- all coexisting at the onset of this transition. A description for clustering behavior is developed from first-principles where the reduction of phosphorus bonds to the lattice, coupled with the resulting residual charge density stabilizing vacancy sites, lowers tensile strain. The broader ambiguity of P_nV cluster species in Si:P is addressed with fundamental analysis, having implications on dopant activation behavior, for example, in this foundational material for microelectronics.

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I. INTRODUCTION

Advancements in transistor miniaturization have increased focus on fundamental performance limits of constituent device materials. Scaling inherently increases electrical resistances, thereby hampering transistor performance, including for source–drains comprised of phosphorus-doped silicon (Si:P).[1–4] A frequently reported obstacle for advances in Si:P conductivity is phosphorus deactivation at high doping concentrations attributed to clustering.[3,5–8] Prior literature has explored strategies to mitigate deactivation, including post-synthesis rapid thermal anneals [e.g., millisecond laser anneal (LA)] and lower temperature growth.[3,5,7–18] For such approaches, subsequent exposure to elevated temperatures (i.e., thermal "soak") inducing deactivation was reported.[9,17–22] One alternative method is post-growth microwave annealing (MWA), which utilizes temperatures of ∼700 °C assisted by electromagnetic radiation in the microwave range to achieve activation.[20,23] This technique was shown to better resist deactivation from subsequent thermal soak (up to 675 °C) compared to lamp-based millisecond anneal (MSA) treatments. Microwave radiation also has high penetration depth to anneal through thick multilayer samples,[24] providing flexibility for process flow integration. This concept hinges on the existence of P_nV defects that form via the diffusion of phosphorus, underlining the kinetics for clustering.[25–27]

Simulations have shown that P_4V has a negative formation energy, indicating that it is the thermodynamically favorable configuration if kinetics allow.[6,20,28] This configuration is referred to elsewhere as a Si_3P_4 cluster, denoting the same four-phosphorus, one-vacancy arrangement discussed throughout this work as P_4V.[6,7] This should be distinguished from bulk, extended Si_3P_4 compound phases independently described in the literature, which are structurally and thermodynamically distinct crystalline phases rather than point-defect clusters embedded in a Si host lattice.[24,29] Dhayalan *et al.* used volume-relaxed density functional theory (DFT) to calculate a negative formation energy for this P_4V/Si_3P_4

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configuration and, combined with positron annihilation spectroscopy (PAS) evidence of phosphorus-decorated vacancies, concluded that it was the most probable inactive-phosphorus complex in highly doped Si:P.[6,7] Their analysis did not, however, incorporate pseudomorphic strain boundary conditions or direct comparison to experimental lattice parameters and did not evaluate P_3V as a competing configuration of comparable viability. The present work addresses both gaps: by directly substantiating candidate P_nV structures against experimental high-resolution x-ray diffraction (HR-XRD) lattice parameters under pseudomorphic constraints, and by providing a structural, charge-density-based explanation for why P_3V and P_4V specifically are favored. This extends the formation-energy result of Dhayalan *et al.* with a quantitative structural mechanism and an explicit comparison against a range of candidate cluster sizes. Despite the effectiveness of MWA, it is thought to only impact PV, P_2V, and P_3V since they have nonzero dipole moments.[20] Given reports of P_4V being most favored, further examination of the atomic structures of these clusters is necessary to resolve this apparent discrepancy.

A variety of DFT simulations and experiments with PAS and Rutherford backscattering spectroscopy (RBS) have helped glean some understanding of P_nV clusters. RBS has shown that phosphorus does not deactivate by simply occupying interstitial sites.[30] On the contrary, interstitial phosphorus appeared only after LA, coinciding with activation.[7] DFT has predicted phosphorus atoms prefer to stay separated; however, pairing with vacancies is favored.[7] PAS showed an increase in the vacancy concentration with higher phosphorus doping and a decrease in vacancies after LA, due to P_nV dissociation, explaining the resistivity reduction.[7] With a combination of PAS indicating phosphorus-decorated vacancies and DFT predicting negative formation energy of P_4V, it was deduced that P_4V is the most probable configuration of these complexes.[6,7] However, this conclusion rested on formation energy comparisons alone, without structural or strain-based validation against experimental lattice data, nor consideration of whether P_3V is equally or more viable—gaps the present pseudomorphic framework directly address. Additionally, this deduction regarding P_4V references PAS data that itself detected P_3V as the highest order cluster.[27] At annealing temperatures >400 °C, these PAS data show P_2V converting to P_3V, and any subsequently higher temperature does not demonstrate higher-order phosphorus decoration. Whether this was a consequence simply of the Si:P sample not having enough phosphorus (1×10^{20} cm^{-3} was considered) for higher-order clustering is unclear. There has also been treatment of negatively charged P_nV clusters, which accurately explains Si:P experimental observations of carrier-concentration plateauing and then declining with continued doping.[31–33] Therefore, the atomic structure of electrically inactive phosphorus in Si:P remains unresolved despite extensive experimental and theoretical study. It is necessary to determine which cluster species are realistic, as well as if they are charged given the high Fermi levels involved.

DFT has been useful in studies of P_nV defects.[6,7,20,30] While these results are valuable for conceptual insights, none take into account that epitaxial synthesis occurs pseudomorphically on a Si substrate[3] because they utilize volume relaxation where all three dimensions of the unit cell are degrees of freedom. In coherent epitaxy on Si substrates, the in-plane lattice constants of the unit cell (a, b) are fixed to that of underlying silicon, $a_0 = 0.5431$ nm,[34] while the perpendicular "c" direction becomes strained from the shorter bond length for ionized phosphorus relative to Si.[8,30,35] This issue has impeded direct comparisons of DFT to experimentally determined values of c from HR-XRD, which would otherwise help tie P_nV configurations to experimental data. Figure 1 shows volume-relaxed DFT-based lattice parameters of Si:P consistently underestimate lattice contraction from phosphorus relative to HR-XRD,[30] as expected since the additional in-plane degrees of freedom (a, b) minimize the contraction of c. One study also outlines this issue and uses the Poisson ratio of Si to correct volume-relaxed DFT lattice constants.[3] However, the exact correction method is unclear and, more importantly, the insight was only used to compare calculated HR-XRD peak positions with/without vacancies and interstitials, not direct comparison to experimental data. Here, a first-principles framework that incorporates pseudomorphic boundary conditions directly into calculations is introduced, enabling quantitative comparison to experimental lattice parameters. This assessment is subsequently used to evaluate the physical viability of various P_nV configurations predicted by DFT. The present analysis is based on first-principles calculations and published experimental observations, and does not rely on any specific processes.

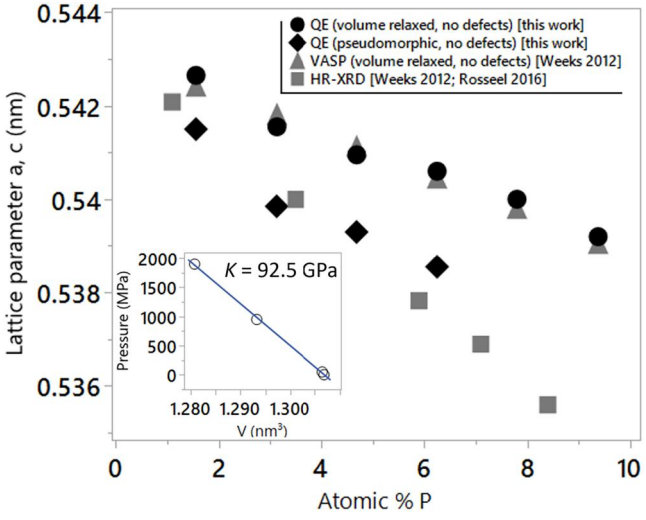


FIG. 1. Comparison of volume-relaxed and pseudomorphic DFT calculations from this study, using Quantum ESPRESSO (QE), to volume-relaxed Vienna *Ab initio* Simulation Package (VASP) DFT of Ref. 30 and HR-XRD of Refs. 3 and 30. Calculations of this 2 × 2 × 2 Si:P system of 64 atoms do not yet consider defects of any kind (i.e., vacancies or P_nV clusters). Volume-relaxed lattice parameters are a (where $a = b = c$), while those for pseudomorphic QE and HR-XRD datasets are c, which is perpendicular to $a = b = a_{Si}$. Inset: Bulk modulus, K, verification of intrinsic Si based on volume-relaxed QE.

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II. COMPUTATIONAL METHODS

Ab initio calculations were conducted with Quantum ESPRESSO (QE)[36] and charge density visualizations utilized VESTA.[37] Baseline runs used to reproduce volume-relaxed results from Ref. 30 for cross-package verification used a $2 \times 2 \times 2$ 64-site diamond cubic lattice with Projector-Augmented Wave (PAW) pseudopotentials,[38,39] Perdew–Burke–Ernzerhof (PBE) exchange-correlation function,[40] 360 eV plane wave cutoff, a $2 \times 2 \times 2$ Monkhorst–Pack K-point grid for Brillouin zone sampling,[41] Methfessel–Paxton electronic smearing of 0.05 eV,[42] collinear spin polarization, and a Broyden–Fletcher–Goldfarb–Shanno (BFGS) ion optimization limit of 0.5 eV/nm.[43] For pseudomorphic calculations, relaxation degrees of freedom was changed to single-axis ($\bar{c}$). Pseudomorphic calculations for comparison to experimental lattice parameters used a larger $3 \times 3 \times 3$ 216-site diamond cubic lattice, for improved flexibility in simultaneously matching total and active phosphorus amounts, and gamma-point Brillouin sampling. Lattice-free Kinetic Monte Carlo (KMC) was implemented using a Stochastic Simulation Algorithm (SSA)[44] with NumPy for probabilistic sampling[45] in Python 3.13.5.

III. RESULTS

A. Defect-free baseline calculations and accuracy improvement from pseudomorphic treatment

As previously mentioned, Fig. 1 shows that Vienna *Ab initio* Simulation Package (VASP) DFT-based volume-relaxed lattice parameters consistently underestimate lattice contraction due to phosphorus doping vs experimental HR-XRD.[30] Since current work uses QE, the same conditions were used to verify that calculations agree between packages. Utilizing the same scaling procedure from Ref. 30, consequently carried out throughout this study, the values, indeed, agree. Additionally, an intrinsic and defect-free Si lattice constant of 0.546 65 nm and Si bulk modulus of $K = 92.5$ GPa were obtained, in reasonable agreement with 0.5464 nm and 93.6 GPa of Ref. 30 and experimental values of $a_{Si} = 0.5431$ nm[34] and $K = 98$ GPa.[46] The impact of switching from volume-relaxed calculations to pseudomorphic is also provided. Values of c are in better agreement with experimental HR-XRD across the atomic % P range as fixing lattice parameters a and b translate to more strain in c, the perpendicular direction, thus resulting in relatively lower values. While these results justify using pseudomorphic methods, Si:P models are still not complete without considering defects.

B. Implementation of P_nV cluster defects

Example illustrations of atomic arrangements for P_nV clusters in Si:P for $n = 2$ and 4 are provided in Fig. 2. Phosphorus atoms decorate the center vacancy in tetrahedral positions. Configurations for $n = 1$ and 3 are similar in that the general tetrahedral geometry persists regardless of the number of phosphorus atoms. Both calculations shown in Fig. 2 are for Si:P with ~7% P–20% of which is "active," a value that will be subsequently explained. Active phosphorus from a DFT perspective is taken to be substitutional and also not clustered in P_nV as this is the only way phosphorus can have a coordination number of four to develop an ionization level of ~45 meV below the conduction band of Si,[47,48] allowing an electron to be donated. Post-calculation atomic positions will adjust due to factors such as void space created by vacancies and the shorter Si–P bond length relative to Si–Si.[35] It is seen in these ~7% P models that should clustering occur with P_2V, lattice distortion is noticeably more than with P_4V. Example localized measurements of Si(001) plane spacing signals more lattice contraction with P_2V than P_4V despite the same levels of total and active phosphorus. This correlates with the additional lattice distortion, as will be subsequently shown. Electron density isosurface images in Fig. 2 illustrate that phosphorus atoms in P_nV clusters have charge lobes where the fourth bond would otherwise be, a structure resembling NH_3 with its lone pair electrons. Since the lobes point into the void, it may

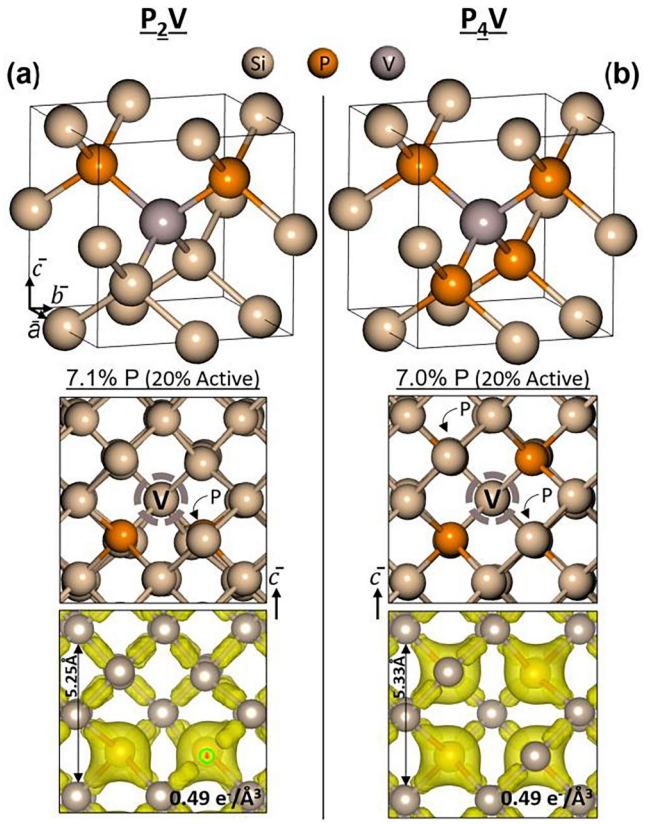


FIG. 2. (a) Top: Example P_2V arrangement in a diamond cubic Si lattice. Coordination to the vacancy is maintained for visual clarity, while, in reality, there is no bonding. Middle: Post-calculation atomic positions for the pseudomorphic Si:P simulation of 7.1% P with 20% active P using P_2V as the clustering defect. The vacancy is behind the visible Si atom. Bottom: Electron density isosurface of 0.49 $e^-/Å^3$ for the corresponding simulation. An example measurement of local Si(001) atomic spacing is included (not meant to represent the broader lattice parameter). (b) Same as (a), but for P_4V clustering and a similar doping level of 7.0% P with 20% active P. Calculations are with a $3 \times 3 \times 3$ system of 216 diamond cubic lattice sites. Axis vector, $\bar{c}$, is perpendicular to the pseudomorphic plane.

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help explain why P_4V Si(001) spacing is not as contracted. For a proper analysis, the broader lattice parameters themselves must first be confirmed and utilized to identify which cluster types are realistic.

C. Comparison of calculations to experiments

The dependence of DFT-based lattice parameters, *c*, on phosphorus doping levels is shown in Fig. 3 for a variety of Si:P defect types and active phosphorus amounts. Included are published experimental values sourced from HR-XRD with various synthesis temperatures for comparison. Temperature impacts % active P, as discussed earlier, and is also shown in dashed/dotted lines of Fig. 3(b) likewise obtained from published experiments.[7,12,13,17] While this aspect is especially important for electrical performance, HR-XRD lattice parameters across all temperature ranges appear to fall on the same linear trend line and, thus, are independent of % active P. DFT confirms fully active Si:P, as well as conditions where clustering occurs via P_3V and P_4V—up to fully inactive—follow the same trend line across all phosphorus concentrations. Therefore, these two cluster configurations are viable. It is important to note that lattice-parameter agreement is a necessary but not sufficient condition: multiple distinct atomic configurations can, in principle, produce indistinguishable values of *c*. This metric, therefore, functions here as a screening criterion that eliminates physically inconsistent configurations (PV and P_2V at high doping) rather than as a unique fingerprint of P_3V/P_4V. Sections III D 1 and III D 2 are used to build a convergent case from independent evidence—energetic favorability and real-space charge/displacement analysis—rather than relying on HR-XRD matching in isolation. As noted, PV dramatically underestimates lattice parameter and thus is physically inconsistent at the doping levels considered here. For P_2V, at 3.3% P, the DFT result is in agreement with experiments; however, at 7.1% P, it also strongly underestimates. Given the excessive lattice distortion this exact calculation showed in Fig. 2(a), it is reasonable that at 7.1% P the lattice prefers P_3V and/or P_4V instead; the latter of which exhibiting reduced disorder [Fig. 2(b)] and agrees with prior reports of negative formation enthalpy.[6,20,28] Thus far, P_2V is possible at lower doping levels and P_3V/P_4V are feasible across the entire range of doping considered here. A $P_2V \rightarrow P_3V \rightarrow P_4V$ transition also cannot be ruled out with increasingly higher doping[23] and/or temperature to assist diffusion kinetics and is considered shortly.

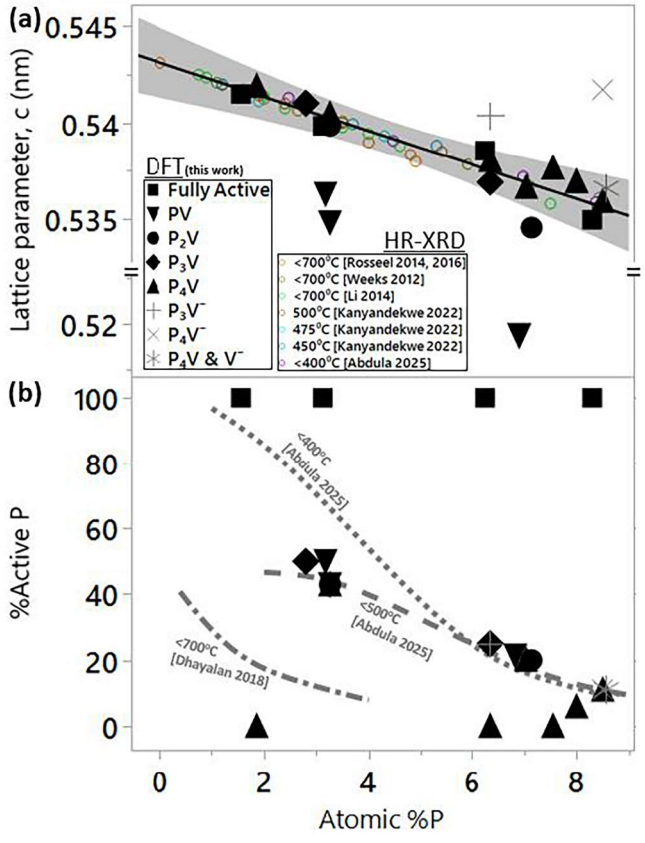


FIG. 3. (a) Filled and line markers: Pseudomorphic DFT-based (QE) lattice parameters, *c*, of Si:P simulated with all phosphorus active (fully active) and various levels of inactive phosphorus via neutral and charged P_nV ($n = 1$–4) clusters. Included is a model with isolated V^- added to P_4V with a concentration of the former as described in the text. Open circles: Experimental values extracted from HR-XRD data of Refs. 3, 11, 16, 17, 30, and 49 for Si:P samples synthesized at various temperatures. Line of fit confidence is $\alpha = 0.05$. (b) Inactive phosphorus levels corresponding to DFT data in (a) and experimentally observed trends (dashed/dotted lines), extracted from Refs. 7 and 17 for three synthesis temperature regimes. Calculations are with a $3 \times 3 \times 3$ system of 216 diamond cubic lattice sites. Simultaneously matching atomic % P and % active P limits what experimentally realistic conditions can be modeled in the given system size.

For clusters that carry a charge, as proposed by Velichko *et al.*,[31,32] these are not physically consistent as seen in Fig. 3 where P_3V^- and P_4V^- overshoot experimental lattice constants. However, negatively charged defects cannot be entirely dismissed as they have accurately modeled carrier-concentration plateau and subsequent decline with increased phosphorus doping in Si:P, a behavior widely reported in experiments.[7,13,15,17,18,33] Non-clustered negatively charged vacancies, V^-, have been experimentally detected in Si:P and predicted to dominate over V^0 upon doping, scaling exponentially with Fermi level and having less than half the formation energy at $10^{20}\,cm^{-3}$ phosphorus.[50–55] A model of P_4V with V^- is also provided in Fig. 3 for a phosphorus concentration of 8.5% P and 11% active P—a combination possible for all temperature regimes shown. The resulting calculated lattice parameter agrees with experimental HR-XRD data and is similar to P_4V alone, implying that V^- can exist in the lattice. It has been reported after a steady mobility decline with increasing phosphorus doping in Si:P, due to increased ionized dopant scattering from P^+,[8,56] a rapid mobility decline commences—coincident with the onset of the carrier-concentration plateau—and then the behavior reverses where mobility steadily rises as additional clustering reduces P^+ population.[13,17] However, after this reversal, mobilities still remained significantly suppressed. One report[17] showed a delta of $\sim 7\,cm^2/V\,s$ between approximately 8.5% P and 2.0% P despite having the same active carrier concentration (i.e., same P^+ population) of $5 \times 10^{20}\,cm^{-3}$, and another study[13]

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exhibited a ~9 $cm^2/V\,s$ difference at 1.5×10^{20} cm^{-3} carrier concentration. With V^- expected to exist in Si:P, as discussed above, and contribute to scattering due to Coulombic interaction, to estimate a concentration for the high-doping regime simulation (represented here by 8.5% P), it was first approximated that V^- has a scattering cross section equal to P^+ as a rough first-order estimate based on Brooks–Herring point-defect scattering.[57] It is then noted an additional $\sim 5 \times 10^{20}$ cm^{-3} P^+ is needed to reduce mobility by 7–9 $cm^2/V\,s$ in the low-%P regime (prior to trend reversal and mobility delta).[5,13,17] Thus, $V^- = 5 \times 10^{20}$ cm^{-3} was used for calculations, which is 1% or two sites in a 216-site system. It is later shown that this exact value is not critical for c as 1.6% V^- also has a minimal impact when combined with P_4V clustering. Mobility trend reversal was also reported to occur at progressively lower carrier concentrations for higher Si:P temperature regimes[13,17] perhaps pointing to V^- being facilitated by clustering itself. Thus, the agreeing lattice constants of DFT with experimental HR-XRD support a potential coexistence of V^- with P_4V. However, all various competing defect configurations in the lattice must be compared on an individual basis to explore which are thermodynamically favorable.

D. Analysis of individual defects

1. Initiation and progression of clustering

As the prior section treats models that were purposely constructed to resemble realistic samples, with respect to total and active doping, models of isolated defects in an otherwise pristine Si lattice are now considered to better understand fundamental energetic favorability. Figure 4 shows the lattice parameters of various individual P_nV cluster defects and defect combinations and how they compare to simple non-clustered doping, "P_n." For the latter, phosphorus atoms were spaced at a minimum of fourth nearest-neighbor (4NN), beyond the threshold they begin to interact, and at 2NN where they can.[6] The analysis as the phosphorus atom count, n, incrementally increases is as follows.

***n* = 0:** Neutral vacancies will not form due to severe tensile strain. The amount is approximately equal to adding three phosphorus atoms to the lattice ($n = 3$, P_3). However, should the vacancy be negatively charged, V^-, lattice parameter recovers to a_{Si}. Note that V^- sits 0.5 eV below the conduction band,[50,54] making high concentrations in intrinsic Si unrealistic at room temperature, but this exemplifies negative charge helping offset strain due to vacancies. It is experimentally known that negatively charged vacancies prevent inward displacement of surrounding atoms, unlike V^0.[58]

***n* = 1:** Neither clustered PV nor separated "P_1 & V^0" are favorable over simply maintaining activated phosphorus, P_1. The essence is that creating a neutral vacancy, clustered or not, is strongly energetically unfavorable. However, if the vacancy is negatively charged, then strain is again fully compensated, as "P_1 & V^-" shows, meaning that V^- can coexist with active dopants. This is realistic since a single phosphorus atom in this 64-site system equates to 1.6% P doping, which is considered degenerate with a Fermi level encroaching the conduction band. With P_1 being ionized (i.e., P^+), it is expected to form PV via Coulombic attraction, yet it is clear that the strain penalty is high. This provides a possible explanation for the discrepancy of DFT with experiments in Fig. 3 for PV and why it is not an optimal configuration of inactive phosphorus at these doping levels.

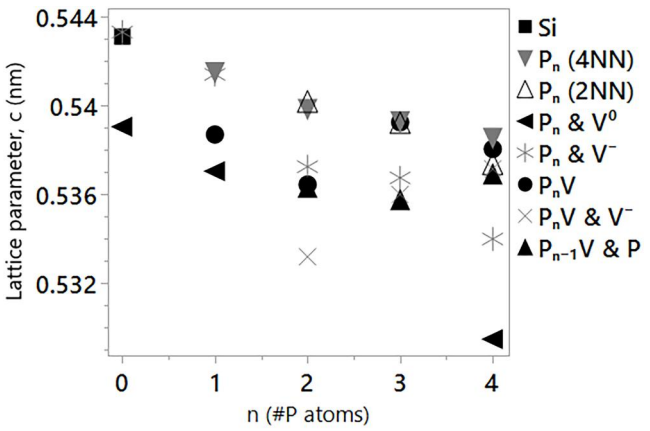


FIG. 4. Impact of individual defects (e.g., P_nV clusters) and defect combinations on lattice parameter, c, from pseudomorphic DFT calculations. Values for active non-clustered P_n doping, where phosphorus atoms are spaced at either a minimum of fourth (4NN) or second (2NN) nearest-neighbor distances, are included for comparison. Also shown are instances where isolated vacancies with a −1 charge (V^-) and neutral (V^0) are added, as well as the reference for pristine Si of 0.5431 nm. A combination of a P_nV cluster plus an isolated 4NN phosphorus ($P_{n-1}V$ & P) is included to assess whether cluster growth is favored. A 2 × 2 × 2 system of 64 diamond cubic lattice sites is used to better accentuate deltas of single defects over the 216-site system. Doping levels for n = 1–4 equate to ~1.6%, 3.2%, 4.8%, and 6.3%, respectively.

***n* = 2**: A simulation of PV with an additional P atom ($P_{2\text{-}1}V$ & P; i.e., "PV & P") does indicate that P_2V is possible should PV ever exist in the first place, thereby inducing cluster growth. As discussed above, however, PV is unlikely. Non-clustered phosphorus also prefers to stay active, regardless of the nearest-neighbor distance (4NN or 2NN); thus, significant levels of P_2V are also not possible at these doping levels. V^- is not likely to accompany active phosphorus (i.e., "P_2 & V^-") in this condition and consistently stays this way with increasing n. Yet, it is clear V^- still offsets some vacancy strain vs V^0, as seen with $n = 4$, but not entirely as with lower n because V^0 induces progressively more strain as doping increases [compare "P_4 & V^0" vs P_4 (4NN) to the same set at $n = 1$].

***n* = 3:** Again should P_2V ever exist in the first place, there would be a thermodynamic tendency to relieve strain by forming P_3V as additional phosphorus is added. Here, however, the first clear indication of a clustering pathway is seen. Active P_3, regardless of nearest-neighbor distance, and P_3V have the same lattice parameter. One DFT study agrees, having calculated a small negative formation energy for P_3V as well.[28] This is reinforced with "P_3 & V^-" and "P_3V & V^-" also having similar lattice parameters. Cluster coexistence with V^- is not yet favored here but does trend toward viable with higher n (see "P_nV & V^-" data points). Regardless, P_3V is a thermodynamically favored cluster path.

***n* = 4:** The lattice-parameter difference between "P_3V & P" and P_4V is small relative to the differences already established for

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lower-n transitions. This indicates a comparatively weak thermodynamic driving force for $P_3V + P \rightarrow P_4V$ conversion, raising the possibility that both species coexist rather than fully converting. It is proposed that this weak driving force reflects a kinetic bottleneck—P_3V is expected to be resistant to dissociation without external stimulus (e.g., LA or MWA)—creating competition between new P_3V nucleation and existing-cluster growth for the same phosphorus supply, a hypothesis examined quantitatively via KMC in Sec. IV B 2. In addition, while P_4 (4NN) marginally having the largest lattice parameter indicates phosphorus would want to reactivate, it cannot since it would leave behind an isolated vacancy. Both a negatively charged vacancy "P_4 & V^-" and neutral vacancy "P_4 & V^0" would induce strong residual strain. Lastly, although there is a penalty for "P_4V & V^-" relative to P_4V alone, it is small and it cannot be ruled out that isolated V^- accompanies P_4V, as discussed earlier. More generally, there are many conditions resulting in similar c, opening the possibility of P_3V, P_4V, active phosphorus, and V^- all coexisting, which experimental data in Fig. 3 also support. A summary of key defect configurations and outcomes of their analysis is provided in Table I.

2. Structural analysis

a. Displacement behavior of V^0, PV, and P_2V. With no doping, it is seen in Fig. 4 that the neutral vacancy V^0 dramatically reduces c as the surrounding Si atoms can tuck into the void, as demonstrated in Fig. 5(a). The effect is so dominant that the addition of one phosphorus atom to form PV does not significantly change c, despite P_1 reconfirming that phosphorus further contracts an otherwise pristine Si lattice. The phosphorus atom in PV does not drift into the void, which would equate to a higher value of c relative to isolated V^0. However, the shorter Si–P bond length[35] induces tensile strain on the surrounding three Si atoms to an extent that neutralizes the lack of inward drift by the P atom, thereby the two effects cancel each other with respect to the lattice parameter, as shown in Fig. 4. While similar c may again be expected for P_2V based on this explanation, a smaller c results, consistent with stronger displacement of the remaining two Si atoms in this cluster configuration [see Fig. 5(b) inset]. Si–P bonds are less prone to distortion to assist in energy minimization, whereas Si–Si bonds are more compliant.

b. Doping-dependence of P_2V viability. The P_2V condition in Figs. 4 and 5 corresponds to a doping of 3.2% P, very nearly the 3.3% P value of Fig. 3 with 43% active P. The former, however, has 0% active P, shows marked lattice distortion, and has a drastically smaller lattice parameter than P_2 (fully active). The latter shows agreement with experiments as well as with DFT of fully active models. Coupled with the fact that at 7.1% P with 20% active P, the P_2V lattice parameter was shown to over-contract vs experiments, it is clear that P_2V, indeed, becomes less plausible with lower % active P and, consequently, higher atomic % P, as Fig. 3 established. Figures 4 and 5 extend this finding by identifying its structural origin: P_2V not fully compensating for strain due to vacancies, which compounds as doping rises. The experimental agreement for P_2V at 3.3% P in Fig. 3 may be due to doping not yet being enough to show clear separation from the fully active and P_3V/P_4V trends.

3. Convergence of P_3V/P_4V with fully active P_n

Another key point Fig. 4 provides is P_3V and P_4V (i.e., 0% active P) lattice parameters closely resemble P_3 and P_4 (i.e., fully active), respectively, thus clarifying why all DFT-based lattice parameters for these two P_nV clusters and fully active cases align across the entire range of atomic %P in Fig. 3, regardless of % active P. A prior volume-relaxed DFT study also reported matching lattice constants for P_4V and P_4;[7] however, P_3V was not considered nor were any data directly compared with experiments. For $n = 3$ and 4, DFT-computed lattice parameters here are indistinguishable from vacancy-free P_n at equal doping (Fig. 4). This indicates the strain from V^0 is neutralized with these two cluster types. It is proposed that this neutralization is the thermodynamic driving force for cluster formation. Atomic distortion around the vacancies in P_3V and P_4V are also minimal, as shown in Fig. 5, consistent with the lattice parameter behavior. Minimal distortion includes the single Si atom in P_3V, which is unintuitive. Electron density isosurfaces again point to charge lobes acting as a stabilizing component. To quantify this picture, Fig. 5(b) shows the electron charge in the void calculated with two methods. First, integrating charge density within a sphere of radius $r = 2.36$ Å centered at the vacancy site (excluding core charge of adjacent atoms). One difficulty with this is convolution with Si atoms protruding into the void for PV and P_2V, which contribute their own valence charge; therefore, only P_3V and P_4V were calculated. The second method compares Bader charge basins[59] of phosphorus atoms in P_nV and P_nSi to determine void charge contribution from

TABLE I. Summary of outcomes from DFT analyses and comparison to experimental HR-XRD values for key defect configurations.

Configuration	HR-XRD agreement	Energetics	Status
PV	No (underestimates c)	Unfavorable vs active P	Excluded
P_2V	Yes at 3.3% P; no at 7.1% P	Unfavorable vs active P	Excluded at high doping
P_3V	Yes, across full range	No penalty vs active P	Retained
P_4V	Yes, across full range	No penalty vs active P	Retained
P_3V^-/P_4V^-	No (overshoots c)	Not evaluated independently	Excluded
P_3V/P_4V/active P/V^- coexistence	Consistent (similar c)	Not independently minimized; KMC-supported (Fig. 6)	Proposed, not uniquely demonstrated

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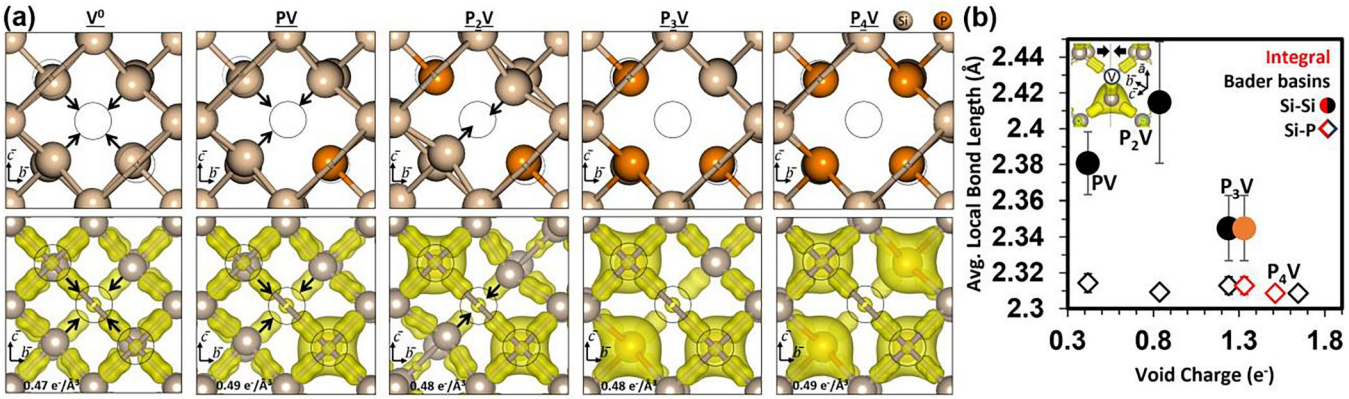


FIG. 5. (a) Top: Resulting atomic positions for the P_nV clusters and V^0 calculated in Fig. 4. Bottom: Electron density isosurfaces of ~0.48 $e^-/Å^3$ for the corresponding simulations. Arrows serve as guides to the eye for atomic displacement directions. Select out-of-plane Si atoms removed for clarity, denoted by black outlines. Axis vector, $\bar{c}$, is perpendicular to the pseudomorphic plane. Calculations are with a 2 × 2 × 2 system of 64 diamond cubic lattice sites. (b) Si–Si and Si–P bond length averages of the first nearest-neighbor atoms to the vacancy for clusters in (a) and their dependence on charge within the void. The latter is determined by two methods: Bader basins for phosphorus atoms in the clusters and an integral method where charge density was integrated over a sphere of $r = 2.36$ Å centered in the vacancy. Longer bond lengths describe atomic displacements further into and around the void. Inset: alternate view of the electron density isosurface for P_2V showing Si atoms displaced along a tetrahedral edge.

phosphorus. Charge values increase with n, and, for P_3V and P_4V, they correlate with a reduction in local bond lengths of atoms immediately surrounding the vacancy, indicating suppressed drift into and around the vacancy. Similar behavior of Si atoms avoiding relaxation into a vacancy due to charge has been experimentally reported for V^- vs V^0.[58] For P_2V, movement occurs along a tetrahedral edge with the two Si atoms approaching each other [Fig. 5(b)], thereby maintaining elevated bond length. With P_3V, this last degree of freedom is removed as all tetrahedral edges for the Si atom to move along then end with a charge lobe, thereby inducing a stronger decrease in bond length, agreeing with the abrupt jump in c (Fig. 4) as atoms cannot relax into the vacancy. Calculations here indicate that phosphorus charge lobes of P_nV are stabilizing factors against lattice distortion for $n = 3$ and 4 where they negate vacancy-induced strain, as shown in Fig. 4. This provides a fundamental framework supporting P_3V and P_4V being key clusters in highly doped Si:P.

IV. DISCUSSION

A. Low doping levels

It is apparent PV and P_2V are not predominant cluster types in the >1.5% P doping range studied here, while P_3V and P_4V are supported by experiment and calculations. However, it is beneficial to consider what occurs at low doping levels since lower-order clusters have been experimentally verified. PAS studies on 3×10^{16} cm^{-3} Si:P showed that only PV forms, which immediately dissociates, with annealing above *room temperature*.[27] This agrees with results described in Fig. 4, where PV clusters are not expected to be stable. Reference 27 further tested Si:P with a higher doping of 1×10^{20} cm^{-3} (0.2% P) and reports PV → P_2V at >120 °C and P_2V → P_3V at >400 °C with no indication of P_4V formation up to the max temperature of ~900 °C. Given that clustering is a diffusion-based mechanism—dependent on both thermal energy and concentration—these PAS data further substantiate P_3V being the first cluster to broadly form with the higher doping levels of >1.5% P considered here. This also highlights that if phosphorus is not in abundant supply, lower-order P_nV is possible, while higher orders (i.e., P_4V) are not. This does not necessarily contradict results in Fig. 4, where PV and P_2V are not favored, because lattice parameter is a broader property tied to total system energy. With small amounts of PV and P_2V that would exist at 0.2% P, an impact on c is less significant than if every atom of a 1.5% P sample turned into PV and at 3.2% P into P_2V. Thus, alternate driving forces to cluster, such as Coulombic attraction between P^+ and V^-, may well be realized locally and in small amounts before strain compounds at higher phosphorus levels, making it less probable.

B. Higher doping levels and coexistence of defect types

1. MWA activation and $P_3V/P_4V/V^-$ coexistence

With strain penalties for PV and P_2V, forming P_3V directly is favored given an abundance of phosphorus. Furthermore, as shown in Fig. 4, the P_3V → P_4V transition appears could be driven by dopant availability and kinetics rather than strong energetic preference (unlike PV → P_2V → P_3V) since both cluster types do well to abate strain due to vacancies. Along with a competition for phosphorus between forming additional P_3V and growing existing clusters to P_4V, the coexistence of the two configurations is also consistent with MWA, demonstrating benefits at 6% P.[20,23] Due to the lack of dipole moment from P_4V, improvements seen imply a significant level of P_3V despite the high level of doping. DFT

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results from Figs. 3 and 4 support this, indicating experimental agreement at 6.4% P for P_3V and energetic favorability on an individual cluster basis. Coexistence of P_3V, P_4V, active phosphorus, and V^- at $n = 4$ produces a similar lattice parameter, c, to any of these configurations alone (Fig. 4), and this value is consistent with the experimental HR-XRD trend (Fig. 3). Because HR-XRD cannot distinguish between these configurations, the lattice-parameter data alone cannot rule out simultaneous coexistence of all four species. This coexistence may explain separately reported experimental trends—the carrier-concentration plateau/decline and a sharp mobility drop—by tentatively associating P_3V nucleation, P_4V formation, and V^- appearance, respectively.[7,13,15,17,18,31,32] An experimentally reported shift from a regime where both temperature and phosphorus concentration impact deactivation to being solely concentration-driven[17] may reflect a phosphorus-limited $P_3V \rightarrow P_4V$ transition. The reported mobility drop/reversal[13,17] could be consistent with V^- forming during this transition as some phosphorus would be consumed by cluster growth, leaving less to getter vacancies into P_3V.

2. Kinetic Monte Carlo model

A KMC simulation is presented in Fig. 6, showing 11 residual vacancies for a hypothetical system of 300 initial P_3V clusters and 100 initial vacancies, despite 600 phosphorus atoms available to fully convert all initial clusters to P_4V and all vacancies to P_3V. These results suggest $P_3V \rightarrow P_4V$ is accompanied by residual V^-. In addition, HR-XRD continues showing smooth progression with doping (Fig. 3) since "P_3V & P," P_4, P_4V, and "P_4V & V^-" lattice parameters are all similar (Fig. 4). The KMC model presented here makes several simplifying assumptions whose influence on the conclusions should be made explicit. Equal rate constants ($k = 0.1$) were assigned to both $P_3V \rightarrow P_4V$ growth and $3P + V \rightarrow P_3V$ nucleation. Relaxing this assumption—for example, if $P_3V \rightarrow P_4V$ proceeds faster due to lower activation energy, as its charge-lobe environment is already partially established—would shift the $P_3V/P_4V/V^-$ population balance toward P_4V at a given phosphorus budget, without changing the qualitative conclusion that vacancies persist under phosphorus-limited conditions. Conversely, if nucleation is favored, more V would be consumed at lower starting P, extending the phosphorus-limited regime. The model also does not include continuous vacancy generation during growth, which—as noted above—would sustain a phosphorus-limited regime indefinitely rather than resolving by ~700 starting P atoms. These simplifications mean that the specific residual-V count (11 at 600 starting P atoms) should be interpreted as illustrative of a general competition effect rather than a quantitative prediction for a specific experimental system.

C. Origin of clustering

A key observation of Fig. 4 is that there is no P_nV condition that provides less strain than fully active P_n at any given n. Thus, it remains a question what initiates clustering in the first place. It was reported in Ref. 7 that as active phosphorus atoms near each other, system energy rises, supposedly due to the repulsion of P^+. However, energy decreases when phosphorus approaches a neutral vacancy. Reference 7 attributed clustering to local strain relaxation,

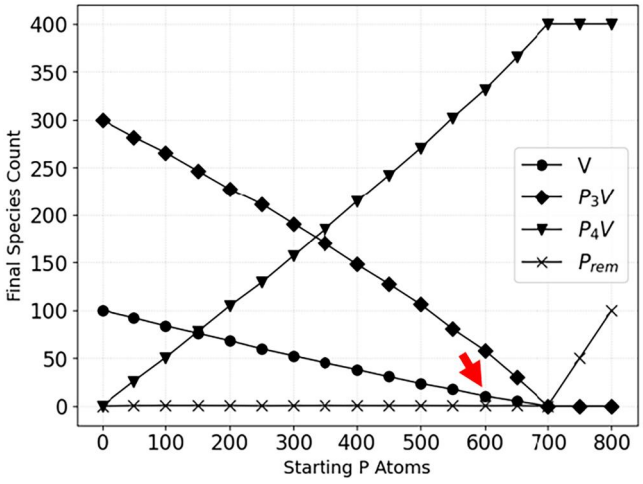


FIG. 6. Kinetic Monte Carlo (KMC) model simulating competing processes for phosphorus: $3P + V \rightarrow P_3V$ and $P_3V + P \rightarrow P_4V$. Resulting P_3V, P_4V, V, and remaining phosphorus (P_{rem}) counts are shown with increasing initial phosphorus atoms. Lower-order cluster formation is assumed to be energetically unfavorable, as discussed in the text, and is bypassed. Incorporation events were assigned equal probability weights ($k = 0.1$) as a worst-case scenario for residual vacancies. Each data point is an average of 100 simulations. Starting populations are $P_3V = 300$ and $V = 100$, effectively requiring 600 P atoms to convert all initial P_3V to P_4V and cluster all V into P_3V. Red arrow highlights residual $V = 11$ at 600 starting P atoms from some V clustering all the way to P_4V, emphasizing the competition for phosphorus. The system stays in a phosphorus-limited regime until 700 starting P atoms, whereas, in reality, creation of new vacancies with more phosphorus (as discussed in the text) sustains this regime.

citing a Si–P bond length decrease from 2.34 (active) to 2.30 Å (clustered). However, the equilibrium vacancy concentration in pristine Si is only $\sim 10^{16}$ cm^{-3} even at 1300 °C[55]—orders of magnitude below the phosphorus doping levels considered here—meaning that this mechanism alone cannot supply enough vacancies from thermal equilibrium. Because deactivation continues to increase with doping [Fig. 3(b)], it is proposed that vacancy formation must be continuously replenished by the doping process itself, rather than drawn from a fixed equilibrium population. This interpretation motivates the strain-relief argument developed here. Doping causes progressively higher tensile strain from shorter Si–P bonds.[35] A vacancy can be introduced and phosphorus clusters around it via P_3V/P_4V as a way to relieve overall strain energy. This is counter-intuitive since creating a vacancy also induces tensile strain. However, with clustered phosphorus, each P atom has one less bond to the rest of the lattice. For example, in Fig. 4, comparable c between V^0 and PV, as well as "P & V^0" and P_2V, indicates clustering with a vacancy helps relieve some strain due to phosphorus. One anchor of the tetragonal geometry is lost and cannot impart strain. Coupled with charge lobes stabilizing bonding in vicinity atoms, as discussed, it is advantageous overall to create such defects. While Fig. 4 does not explicitly show this when comparing P_3 (4NN or 2NN) vs P_3V, it is evident when "P_3 & V^-" is considered, which does have a strain penalty compared

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to clustering. This is consistent with (1) V^- population exponentially increasing with Fermi level,[50,53,54] making it certain to exist in this charge state at degenerate doping levels of P_3 (4.8% P), (2) vacancy formation energy being 2.5× lower in 10^{20} cm^{-3} Si:P than intrinsic when phosphorus atoms are in the vicinity,[55] and (3) PAS data showing more vacancies with higher phosphorus concentration.[7] The principle carries to P_4V as well. Similar experimental behaviors were reported with arsenic, which also imparts tensile strain in Si.[18] Taken together, the energetic comparisons in Fig. 4 and the structural charge-lobe analysis in Fig. 5 are consistent with a proposed physical picture in which high phosphorus doping itself—through the tensile strain it introduces—is the underlying driving force for P_nV clustering, with strain-energy relaxation as the operative premise.

D. Summary of evidentiary strength

Revisiting the summary in Table I, of which individual configurations are supported by direct lattice-parameter and energetic evidence, it is useful to explicitly grade the confidence associated with the dynamic clustering scenarios developed. The exclusion of PV and P_2V (at high doping) as dominant species, and the identification of P_3V as the initial clustering step, rests directly on the lattice parameter and energetic comparisons in Table I and Figs. 3 and 4. The $P_3V \rightarrow P_4V$ transition being kinetically rather than thermodynamically controlled, and the coexistence of P_3V, P_4V, active P, and V^-, are physical scenarios consistent with the available lattice-parameter and KMC evidence, but not uniquely demonstrated by them.

V. CONCLUSION

Pseudomorphic DFT calculations were carried out for a direct comparison to experimentally extracted lattice parameters from published HR-XRD data, which facilitated an analysis of viable defect types in highly doped Si:P. It was shown that P_2V at a doping of ∼7% P and PV do not agree with experiments, thereby making these clusters unlikely. P_2V at a lower doping of ∼3% P and P_3V/P_4V across the entire ∼1.5%–9% P doping range considered here do agree with experiments. Additional calculations of individual defects confirm that PV and P_2V are energetically unfavorable due to strain-energy penalties, whereas P_3V and P_4V do not have such penalties. Upon the structural analysis of DFT results, it is evident that charge lobes from missing bonds of P atoms in P_3V/P_4V negate tensile strain of vacancies, whereas for PV and P_2V, the fewer charge lobes are not sufficient and surrounding Si atoms reposition about the vacancy. Therefore, P_3V and P_4V are the clustering configurations consistent with data. It is proposed tensile strain from progressively higher phosphorus concentrations drives clustering as it is alleviated not only by reducing phosphorus bonding to the lattice with vacancy creation but also by the appearance of lattice-stabilizing charge lobes.

Cluster growth for $PV \rightarrow P_2V \rightarrow P_3V$ is energetically favorable, whereas $P_3V \rightarrow P_4V$ is more consistent with dopant availability and kinetics. Upon the onset of the latter transition, P_3V, P_4V, active phosphorus, and V^- all coexisting is in line with calculations and agrees with referenced experiments. KMC shows a competition for phosphorus between $P_3V \rightarrow P_4V$ cluster growth, and nucleating new P_3V helps manifest significant V^-. Models for charged P_3V^- and P_4V^- do not agree with experimental HR-XRD. MWA requires choosing Si:P with appropriate doping levels since at low doping, % active P can already be sufficient, while at high doping, clusters convert from P_3V, which MWA can influence, to P_4V, which it cannot. This study presents a physically consistent description of phosphorus clustering in highly doped Si:P, enabled by the introduction of a new method to connect first-principles calculations to experiments that can be used for any crystalline material.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Daner Abdula: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The DFT data and code that support the findings of this article are available from the corresponding author upon reasonable request.

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